Name: NetID:

**Worksheet 3: Embeddings and Pooling**

We're going to simulate different kinds of pooling on the last hidden states of an embedding model.

Below, you are given last hidden states for an input sequence. You are also given an "attention mask", which is 1 at positions with non-padding tokens and 0 at positions with padding tokens.

Your job is to generate a single embedding from the hidden states using two different strategies:

1. CLS: take the hidden state that corresponds to the CLS token
2. Mean: take the mean of all hidden states *(not counting padding tokens in your calculation!)*

After extracting the CLS and Mean Hidden States, you will pass both of them through the Pooler.

The Pooler is composed of the Pooler Weights matrix and Pooler Bias vector.

To get the Pooled Embeddings for your CLS and Mean Hidden States:

1. matrix multiply Pooler Weights and the Hidden State, then
2. add the bias.

***Hint: if done correctly, every number should be an integer.***

**Token Index**

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 |

**Attention Mask**

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1 | 1 | 1 | 1 | 0 | 0 | 0 | 0 |

**Last Hidden States**

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| -1 | 1 | 3 | 1 | 0 | 4 | 2 | 1 |
| 2 | -1 | 1 | 2 | 1 | 0 | -2 | 1 |
| 1 | 0 | 1 | -2 | -3 | 1 | 3 | 0 |

**Pooler Weights** / **Pooler Bias**

| | | | |
|---|---|---|---|
| 1 | 0 | 1 | [redacted] |
| 0 | 1 | 1 | [redacted] |
| 1 | 0 | 0 | [redacted] |

**CLS Hidden State**

| |
|---|
| -1 |
| 2 |
| 1 |

**Mean Hidden State (with mask!)**

| |
|---|
| 1 |
| 1 |
| 0 |

**CLS Pooled Embedding**

| |
|---|
| -1 |
| 3 |
| 0 |

**Mean Pooled Embedding**

| |
|---|
| 0 |
| 1 |
| 2 |